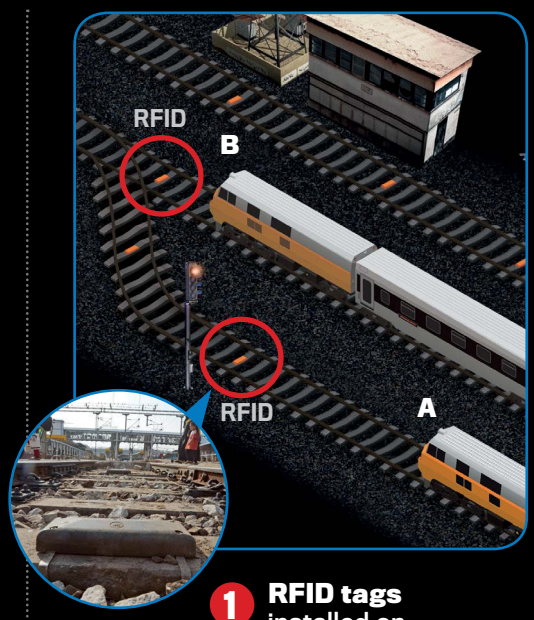
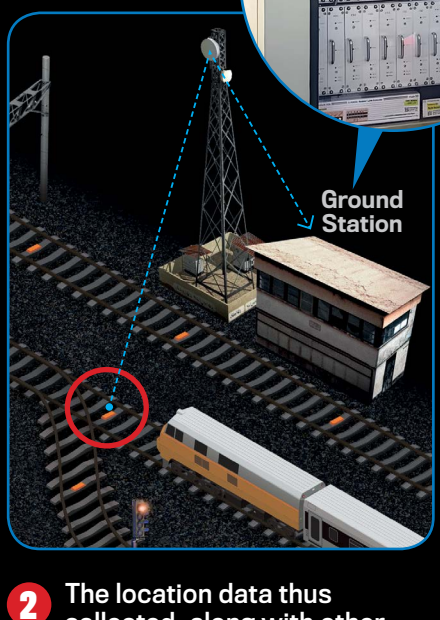
The Kavach network comprises onboard equipment on trains and on-ground station equipment, which communicate continuously via radio frequencies, and Radio Frequency Identification (RFID) tags installed on tracks. Here's how the system works

HOW KAVACH WORKS

India's indigenous Kavach system, first conceptualised in 2012, monitors train safety by facilitating in-cab signalling and sending crucial real-time instructions to the loco pilot. In a dangerous situation—such as when it senses that a driver has misread or is about to jump a signal—it can also automatically apply brakes to avoid collision



1 **RFID tags** installed on railway tracks provide the location of trains A and B



2 The location data thus collected, along with other details like speed, is sent to the **ground station** from the trains



(LTE) [or 4G] frequency. We named it mCOMM. It was a huge achievement," he adds.

An ATP meant to keep the world's fourth-largest railways safe naturally needed rigorous trials. Kavach was gradually deployed in South Central Railway, over 1,400 kilometres and 139 locomotives in sections touching Guntakal, Bidar, Nanded, Manmad, Secunderabad and others. Trials lasted for almost a decade. The success of the measure can be gauged from the fact that there has been no case of overshooting of signals at these Kavach-ready sections.

"The testing has been rigorous," says Srinivas Reddy, director of the Hyderabad-based Medha Servo Drives, the technology partner of the railways. "The security built into it is robust because in this even the machines exchange OTPs among themselves to maintain the integrity of signals. No one can tamper with it." Medha was the first private player to tie up with the railways' Research Design and Standards Organisation (RDSO) for Kavach. Since then, two more Indian firms—Kernex and HBL Power Systems—have joined the league. The RDSO jointly holds the intellectual property with them. Four more firms are developing their own systems—GGtronics, Quadrant FutureTek, Areca and the state-owned BHEL—while Japan's Kyosan, which supplies signalling systems to the railways, and Germany's Siemens, are also working on Kavach, say officials.

In 2018, Europe's Italcertifer awarded Kavach its Safety Integrity Level-4 (SIL-4) certificate, the highest level in the standard developed by Geneva's International Electrotechnical Commission to assess the probability of a safety system allowing a hazardous error. The certification puts Kavach on par with ATPs in developed railway systems, like that in Europe. The railways, in fact, says the certification implies that Kavach cuts the probability of dangerous errors to once in 10,000 years.

Costs, too, may go down as the programme is scaled up. "In the early development stages, a lot of equipment used to be imported. Now, we manufacture



"WE HAVE TAKEN UP KAVACH ON A MISSION MODE. THE WAY RAILWAY ELECTRIFICATION IS HAPPENING, KAVACH WILL BE INSTALLED IN THE SAME WAY"

ASHWINI VAISHNAW, *Minister for Railways*